

# Mohammed Elghanam

## R&D CTO | Embedded & IoT Product Development

Technical Project Portfolio - Embedded Systems, Smart Utilities, LPWAN IoT, FPGA, RTU, Power Protection, Medical Control Systems & Safety-Critical Platforms

This portfolio presents selected R&D and engineering systems led, architected, supervised, reviewed, or contributed to across smart utilities, IoT, industrial automation, power systems, medical control, communication platforms, and safety-critical environments.

Designed for executive review, technical due diligence, senior hiring, consulting, partnerships, and technology collaboration.

**25+**

Years R&D leadership

**5**

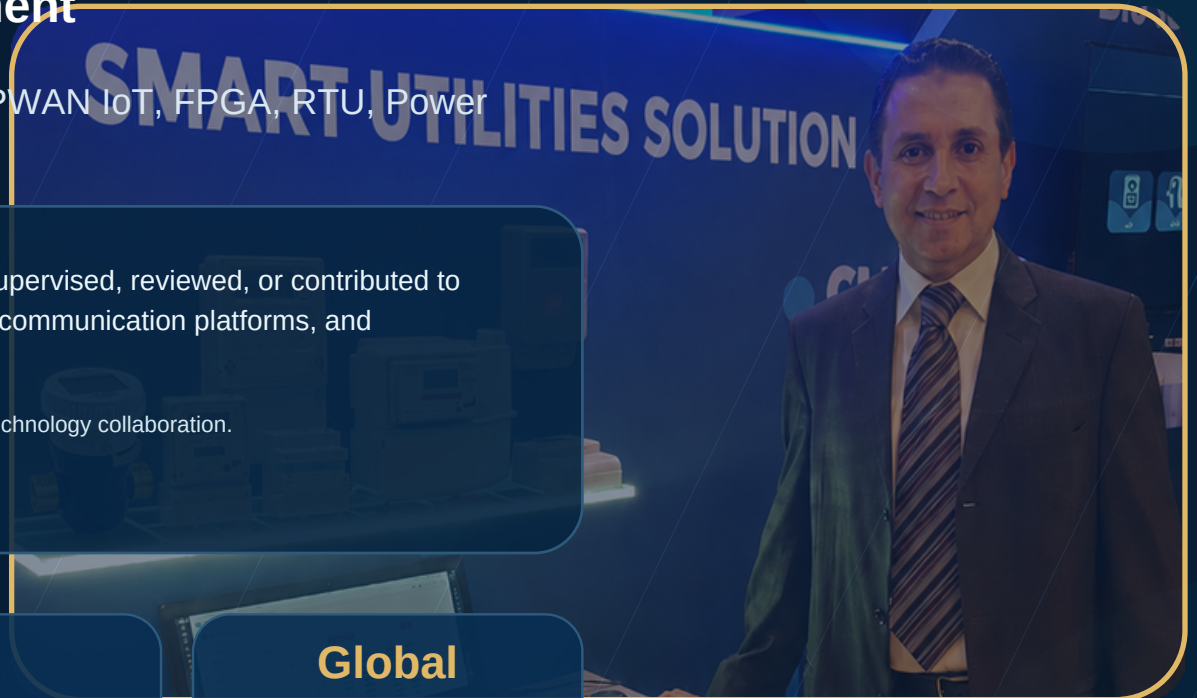
Granted patents

**9**

Project case studies

**Global**

Executive positioning



# Project Portfolio Architecture

Selected systems reframed as executive project case studies - each page highlights the problem, engineered solution, role, architecture lens and enabling technologies.

9

Project case studies

6

Engineering domains

25+

Years of R&amp;D leadership

5

Granted patents

01

## Smart Utilities

Metering, AMI, LPWAN, PLC, billing and data integration

02

## Industrial Control

RTU, EMS, industrial protocols, validation and deployment

03

## Power Protection

Digital relay, 11kV protection, commissioning and DCS

04

## FPGA & SDR

High-speed processing, ARM/FPGA/DDR3 and SI/EMC review

05

## Medical Control

Ventilator and oxygen concentrator control-system R&D

06

## Safety-Critical

Railway control, safety interlocks, CENELEC-aware thinking

### ROLE PATTERN



Technology strategy



System architecture



HW/FW integration



Validation readiness



Team leadership

# 01 Smart & Prepaid Metering Systems

SMART UTILITIES | ELECTRICITY, WATER & GAS

Modernizing utility metering into connected, data-driven infrastructure.

## ENGINEERING CHALLENGE

Utilities need reliable field metering, flexible communication, prepaid control, billing integration, and scalable data acquisition across electricity, water, and gas networks.

## ENGINEERED SOLUTION

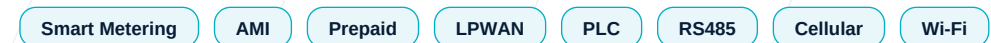
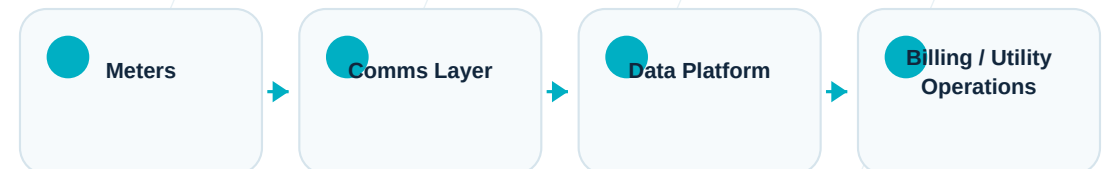
Led development of smart and prepaid metering platforms integrating embedded metering hardware, AMI concepts, LPWAN, RS485, cellular, Wi-Fi and PLC communication with scalable data and billing ecosystems.

## MY ROLE

Technology leadership, hardware architecture, communication technology selection, firmware and software integration supervision, engineering review, validation direction, and product-readiness coordination.



Architecture Lens



## 02 Patented LPWAN IoT Platform

IOT | SMART CITY | SMART UTILITY APPLICATIONS

A low-power, long-range IoT foundation for distributed smart infrastructure.

### ENGINEERING CHALLENGE

Smart-city and utility applications require cost-effective long-range connectivity for distributed devices with low power consumption and robust field coverage.

### ENGINEERED SOLUTION

Spearheaded a patented LPWAN IoT platform from concept to deployment, covering field devices, gateways, embedded software, cloud/mobile integration and applications such as lighting, waste, agriculture and buildings.

### MY ROLE

R&D leadership, end-to-end architecture, embedded device direction, gateway/software integration supervision, IP support, validation and productization guidance.



System architecture visualization

### Architecture Lens



# 03 Advanced RTU / Embedded EMS Platforms

INDUSTRIAL AUTOMATION | MONITORING & CONTROL

Universal industrial control architecture for reliable field intelligence.

ENGINEERING CHALLENGE

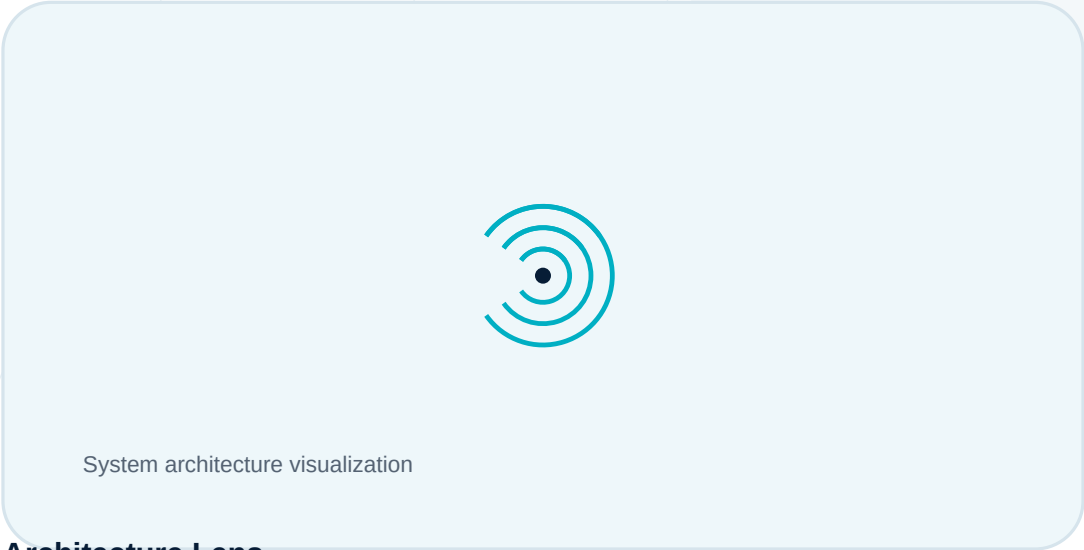
Industrial sites require scalable remote terminal units and embedded EMS functions that integrate hardware, firmware, industrial communication protocols and cybersecurity-aware architecture.

ENGINEERED SOLUTION

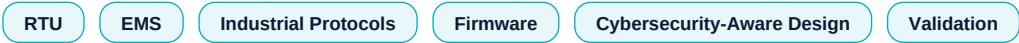
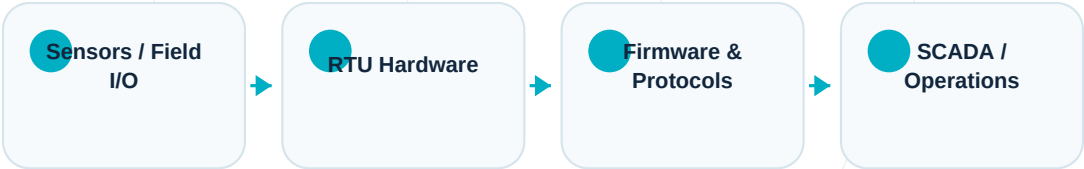
Architected and supervised advanced RTU and embedded EMS platforms for industrial monitoring and control, coordinating hardware, firmware, communication layers, validation and deployment readiness.

MY ROLE

System architecture owner, R&D supervisor, engineering review lead, hardware/firmware integration coordinator, validation and deployment-readiness driver.



Architecture Lens



# 04 Digital Overcurrent & Earth-Fault Relay

POWER GRID PROTECTION | 11KV NETWORKS

Modern protection logic for medium-voltage network reliability.

## ENGINEERING CHALLENGE

11kV networks require reliable fault detection, protection behavior, field installation readiness and integration with existing control systems.

## ENGINEERED SOLUTION

Directed development of digital overcurrent and earth-fault relay systems for 11kV networks, including hardware/software design, rigorous testing, substation installation, commissioning and Foxboro DCS integration.

## MY ROLE

Technical direction, protection-system architecture, hardware/software design supervision, testing coordination, commissioning support and DCS integration oversight.

System architecture visualization

## Architecture Lens



11kV Networks

Protection Relay

Overcurrent

Earth Fault

Testing

Commissioning

# 05 FPGA-Based Mission-Critical Communication Platforms

FPGA | SECURE COMMUNICATION | HIGH-RELIABILITY SYSTEMS

High-speed digital processing for low-latency, secure communication paths

## ENGINEERING CHALLENGE

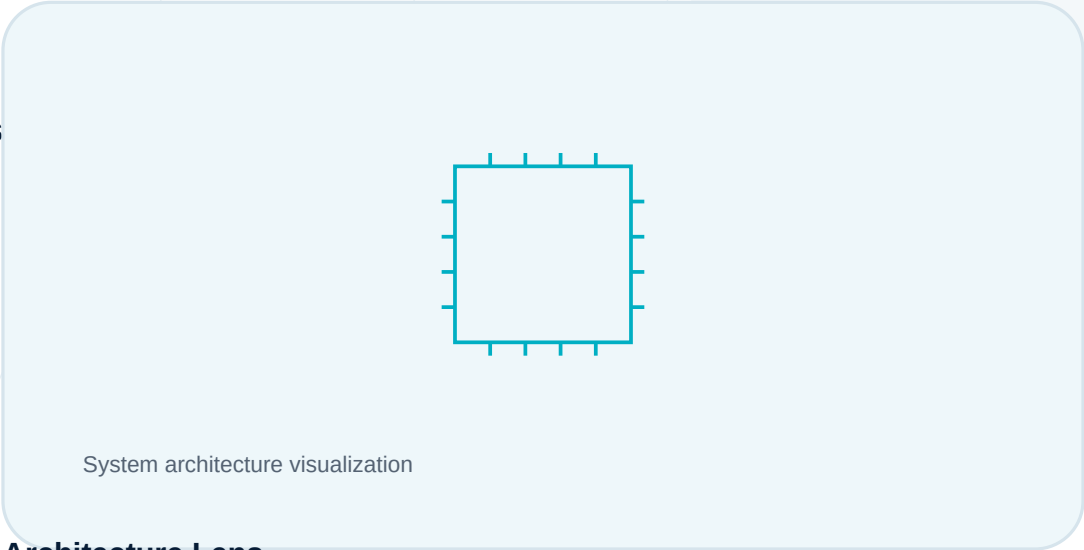
Mission-critical communication platforms require deterministic processing, low latency, optimized FPGA utilization, secure integration and system-level validation.

## ENGINEERED SOLUTION

Supervised FPGA-based communication platforms involving high-speed signal processing, low-latency communication, secure system integration, MMI design, optimized FPGA utilization and validation.

## MY ROLE

R&D supervisor and technical lead for architecture review, FPGA implementation oversight, MMI integration, performance optimization and validation planning.



## Architecture Lens



- FPGA
- VHDL
- Verilog
- Signal Processing
- MMI
- Secure Integration
- Low Latency

## 06 Prototype SDR Platform Hardware Review

SOFTWARE-DEFINED RADIO | ARM + FPGA + RF FRONT-END

Design authority review for high-speed digital and RF-oriented embedded boards

### ENGINEERING CHALLENGE

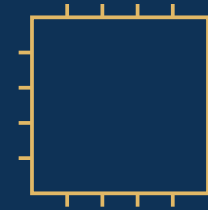
SDR prototypes require careful hardware partitioning, high-speed board discipline, RF-front-end awareness, memory interfaces and signal-integrity/EMC readiness.

### ENGINEERED SOLUTION

Provided hardware design authority for prototype SDR platforms, reviewing ARM processor, Xilinx FPGA, DDR3 RAM, RF-front-end and digital processing boards with SI/EMC guidance.

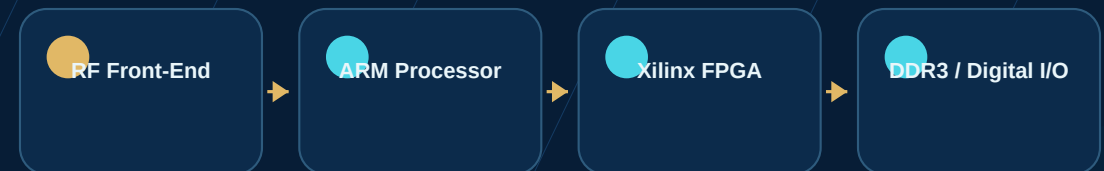
### MY ROLE

Hardware review authority, architecture reviewer, board-level risk identifier, SI/EMC guidance provider and integration-readiness advisor.



System architecture visualization

### Architecture Lens





## 07 Medical Device Control Systems

VENTILATOR & OXYGEN CONCENTRATOR R&D

Embedded control systems developed with safety and calibration readiness

### ENGINEERING CHALLENGE

Respiratory support systems require reliable embedded control, sensor/actuator integration, safety-oriented testing, calibration and regulatory preparation.

### ENGINEERED SOLUTION

Led control-system design and prototyping for ventilator and oxygen concentrator systems. Calibration and safety certification were completed through Cairo University Medical Calibration and Safety Center; Egyptian Drug Authority licensing/production procedures are in progress.

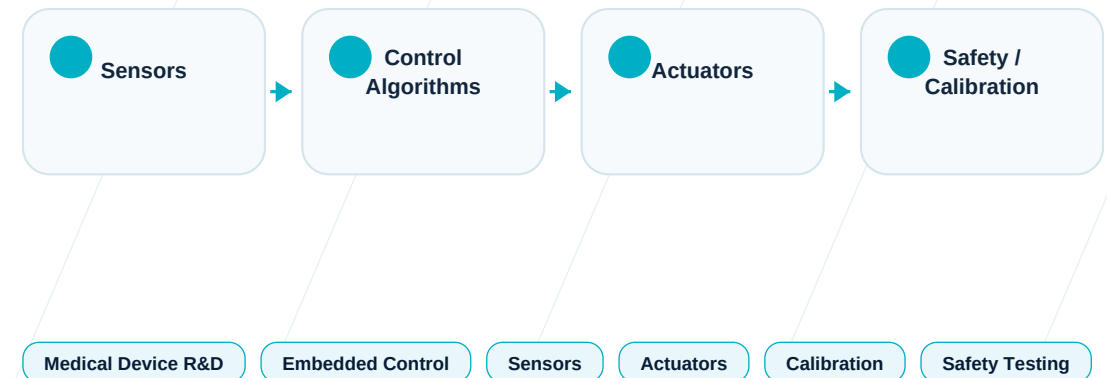
### MY ROLE

R&D lead for control architecture, prototyping direction, embedded control oversight, testing coordination, calibration/safety readiness and regulatory preparation support.



System architecture visualization

### Architecture Lens



## 08 Unified Azan System

GPS TIMING | RDS | LARGE-SCALE SYNCHRONIZATION

Synchronized audio distribution across large urban geographies.

### ENGINEERING CHALLENGE

Large-scale mosque audio synchronization requires accurate timing, resilient communication, signal distribution and integration with existing infrastructure.

### ENGINEERED SOLUTION

Led the technical architecture of the Unified Azan System across Cairo, Qaliubia and Giza using GPS-based timing, RDS, high-power amplification, communication networks and integration with existing mosque systems.

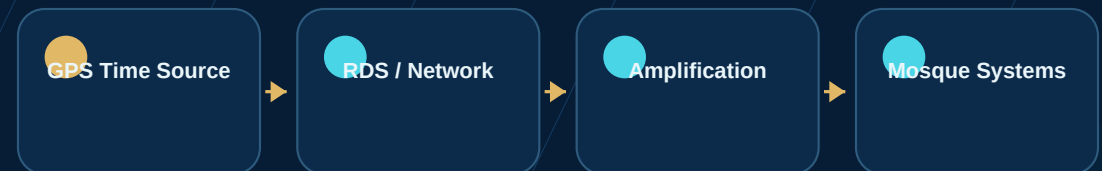
### MY ROLE

Technical architect for synchronization concept, network/audio architecture, timing strategy, system integration and troubleshooting direction.



System architecture visualization

### Architecture Lens



GPS Timing

RDS

Audio Systems

High-Power Amplifiers

Communication Networks

## 09 Railway Level-Crossing Control Systems

SAFETY-CRITICAL RAILWAY CONTROL

Control and safety-interlock contribution within railway infrastructure requirements

### ENGINEERING CHALLENGE

Railway level-crossing systems require disciplined control logic, signal processing, installation readiness, commissioning support and strict safety requirements.

### ENGINEERED SOLUTION

Contributed to railway level-crossing control systems with signal processing units, safety interlocks, installation/commissioning support and adherence to strict safety requirements.

### MY ROLE

Technical contributor supporting control-system elements, signal processing units, safety interlocks, commissioning readiness and safety-aware implementation practices.



System architecture visualization

### Architecture Lens



Railway Control

Safety Interlocks

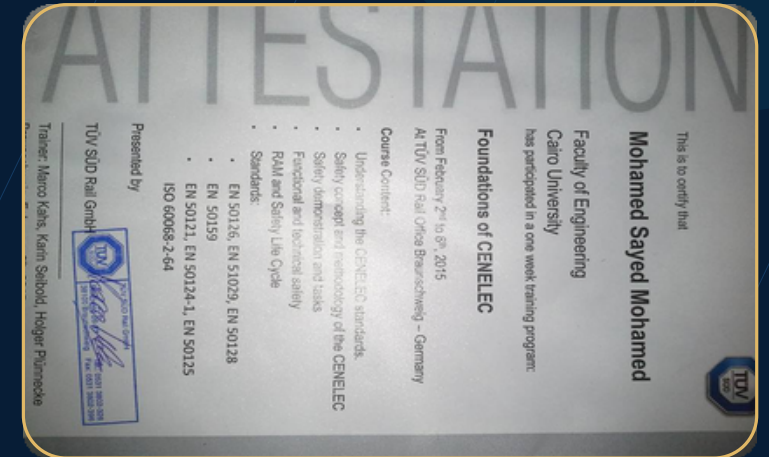
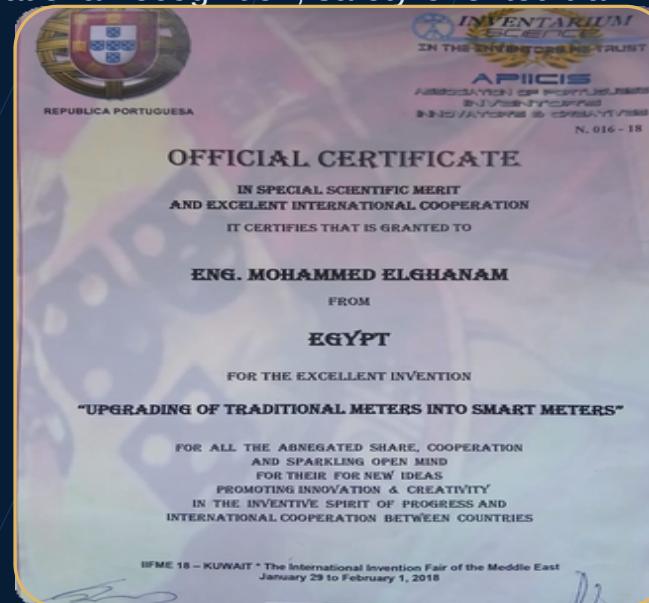
Signal Processing

Commissioning

CENELEC Awareness

# Innovation, Recognition & Engineering Credibility

Evidence supporting the portfolio: patents, international recognition, safety-oriented training, and cross-domain R&D delivery.



## 5 Granted Patents

Related to embedded/IoT and smart metering innovation.

## IIFME Kuwait Recognition

Recognition for upgrading traditional meters into smart meters.

## CENELEC Awareness

TUV SUD Rail training in CENELEC standards, RAM and safety lifecycle.

## Cross-Domain R&D

Utilities, industrial automation, power, medical and communication systems.

# Let's Build What Matters

Available for senior R&D leadership, solution architecture, engineering consulting, embedded/IoT product development, technical advisory, technology due diligence, and productization roadmaps.



## Embedded & IoT Consulting

Architecture, product definition, device/gateway/software integration.



## Smart Utility Digitalization

Metering, AMI, communication layers, data and billing integration.



## FPGA / Hardware Review

FPGA platforms, SDR boards, SI/EMC and production-readiness review.



## Safety-Critical Review

Validation discipline, risk review, CENELEC-aware safety thinking.

## Mohammed Elghanam

R&D CTO | Embedded & IoT Product Development | Engineering Consultant | Technical Expert

Email: [Mohammed.elghanam@gmail.com](mailto:Mohammed.elghanam@gmail.com) | Mobile: +20 01147778583 | LinkedIn: [linkedin.com/in/mohammed-elghanam](https://www.linkedin.com/in/mohammed-elghanam)

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